

Semiconductor Devices

ENGS124 Electricity and Magnetism Lab

13 November 2025

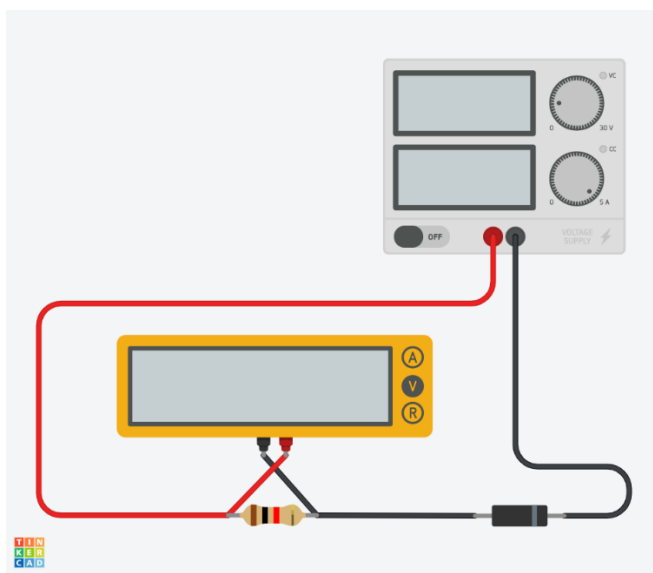
Introduction

Semiconductor devices, such as diodes, thermistors, and Light Dependent Resistors (LDRs), are the core components of nearly all modern electronic systems. Understanding the electrical characteristics of these devices is fundamental to analyzing and designing functional circuits. This experiment is structured into three parts, each designed to investigate the operational principles of a different fundamental semiconductor component. The first part is concentrated on measuring the current-voltage (I-V) characteristic of a diode under both forward and reverse bias conditions and experimentally determining its threshold voltage and conduction behavior. The second part is about LDR, and the third part is about thermistor.

Part 1 Diode I-V Characteristic Measurement

The current-voltage (I-V) characteristics of a diode are measured using a series circuit consisting of a DC voltage source, a current-limiting resistor (R), and the diode under test (D). The circuit configuration utilized for this experiment is shown below:

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Diode	1	A specific semiconductor under test
Resistors	3	A set of 1k Ω , 10 k Ω , 100 k Ω
Connection Wires	~	Copper wires
Breadboard	1	A platform for assembling the circuit

Variables

Name	Type	Resolution
V_{all}	Control	0.01 V
V_R	Measure	0.01 V
V_D	Calculate	0.01 V
I	Calculate	1 mA
R	Parameter	100 Ω

Errors

Name	Type	Impact
Voltmeters Internals Resistance	Systematic	When connected in parallel for measuring, the voltmeter can draw a small amount of current, slightly altering the total circuit current and making the measured voltage lower than the theoretical open-circuit value
Power Supply Fluctuations	Random	The output voltage from the DC power supply may not be perfectly stable and fluctuate slightly.
Contact and Wire Resistance	Systematic	The connecting wires and the contact points on the breadboard have a small but non-zero resistance. This can cause the total resistance to be slightly higher than the sum of the component resistors, affecting the I-V readings.

Formula Used to Calculate the Current in the Circuit

$$V_{all} = V_R + V_D$$

$$I = \frac{V_{all} - V_D}{R}$$

Data

R	V_{all}	V_R	V_D	I
R = 100 kΩ	-1.00	0.00	-1.00	0.00E+00
	-0.50	0.00	-0.50	0.00E+00
	-0.25	0.00	-0.25	0.00E+00
	0.62	0.31	0.31	3.07E-06
	1.18	0.84	0.34	8.40E-06
	1.72	1.36	0.36	1.36E-05
	2.22	1.85	0.37	1.85E-05
	2.75	2.38	0.37	2.38E-05
	3.34	2.96	0.38	2.96E-05
	3.84	3.45	0.39	3.45E-05
	4.37	3.97	0.40	3.97E-05
	4.92	4.51	0.41	4.51E-05
	5.43	5.02	0.41	5.02E-05
	5.99	5.59	0.40	5.59E-05
	6.53	6.13	0.40	6.13E-05
	7.00	6.60	0.40	6.60E-05
	7.56	7.15	0.41	7.15E-05
	8.15	7.73	0.42	7.73E-05
	8.62	8.20	0.42	8.20E-05
	9.16	8.74	0.42	8.74E-05
	9.71	9.29	0.42	9.29E-05
	10.21	9.78	0.43	9.78E-05
	10.80	10.38	0.42	1.04E-04
	11.29	10.86	0.43	1.09E-04
	11.86	11.42	0.44	1.14E-04
	12.19	11.76	0.43	1.18E-04
	12.85	12.41	0.44	1.24E-04
	13.43	13.00	0.43	1.30E-04
	13.95	13.50	0.45	1.35E-04

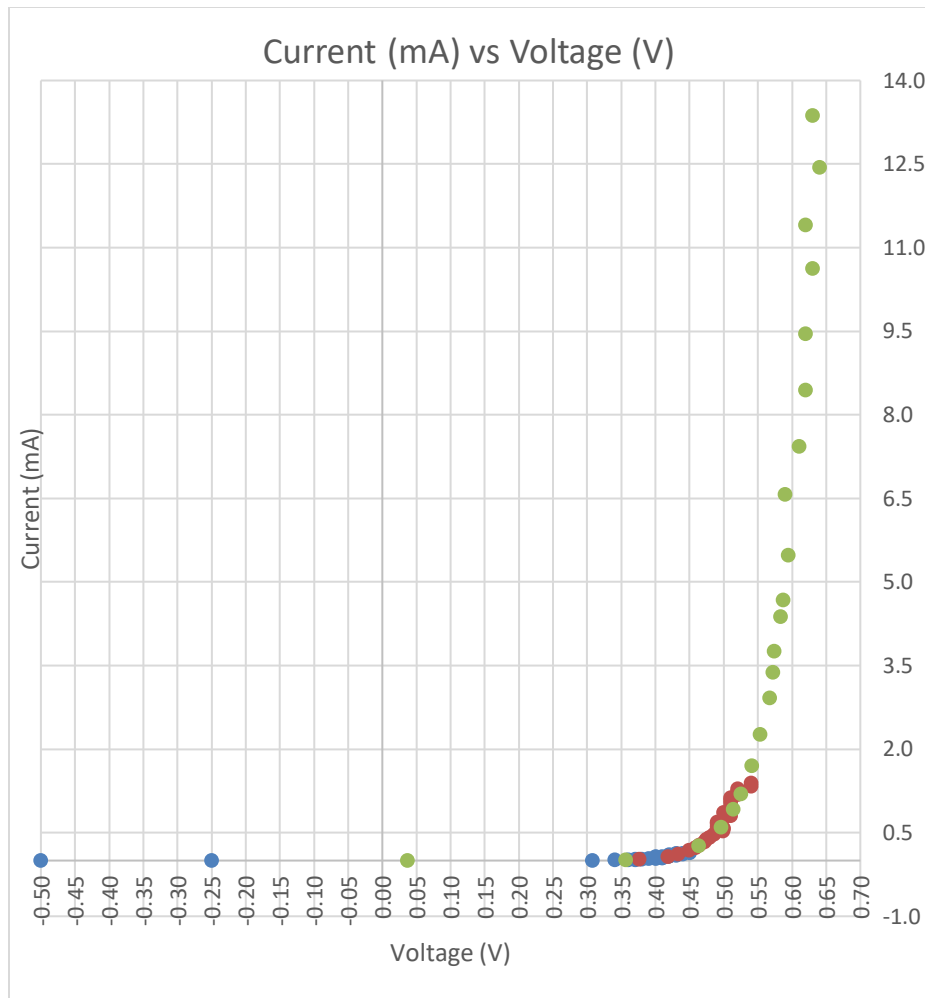
R = 10 kΩ	0.61	0.23	0.38	2.31E- 05
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	1.13	0.71	0.42	7.12E-05
	1.59	1.15	0.43	1.15E-04
	2.28	1.83	0.45	1.83E-04
	2.73	2.28	0.46	2.28E-04
	3.25	2.79	0.46	2.79E-04
	3.79	3.31	0.47	3.31E-04
	4.31	3.83	0.47	3.83E-04
	4.69	4.21	0.48	4.21E-04
	5.26	4.78	0.49	4.78E-04
	5.81	5.31	0.50	5.31E-04
	6.22	5.72	0.50	5.72E-04
	6.57	6.08	0.49	6.08E-04
	7.17	6.68	0.49	6.68E-04
	7.46	6.97	0.49	6.97E-04
	8.05	7.55	0.50	7.55E-04
	8.56	8.05	0.51	8.05E-04
	9.18	8.68	0.50	8.68E-04
	9.69	9.18	0.51	9.18E-04
	10.70	10.19	0.51	1.02E-03
	11.29	10.78	0.51	1.08E-03
	11.82	11.31	0.51	1.13E-03
	12.32	11.80	0.52	1.18E-03
	12.89	12.37	0.52	1.24E-03
	13.41	12.89	0.52	1.29E-03
	13.94	13.40	0.54	1.34E-03

14.43	13.89	0.54	1.39E-03
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R = 1 k Ω	0.04	0.00	0.04	0.00E+00
	0.37	0.01	0.36	1.20E-05
	0.74	0.27	0.46	2.72E-04
	1.09	0.60	0.50	5.96E-04
	1.43	0.92	0.51	9.16E-04
	1.72	1.19	0.53	1.19E-03
	2.25	1.71	0.54	1.71E-03
	2.81	2.26	0.55	2.26E-03
	3.49	2.92	0.57	2.92E-03
	3.95	3.37	0.57	3.37E-03
	4.33	3.76	0.57	3.76E-03
	4.96	4.37	0.58	4.37E-03
	5.27	4.68	0.59	4.68E-03
	6.08	5.48	0.59	5.48E-03
	7.16	6.57	0.59	6.57E-03
	8.04	7.43	0.61	7.43E-03
	9.06	8.44	0.62	8.44E-03
	10.07	9.45	0.62	9.45E-03
	11.26	10.63	0.63	1.06E-02
	12.03	11.41	0.62	1.14E-02
	13.08	12.44	0.64	1.24E-02
	14.00	13.37	0.63	1.34E-02

Plot

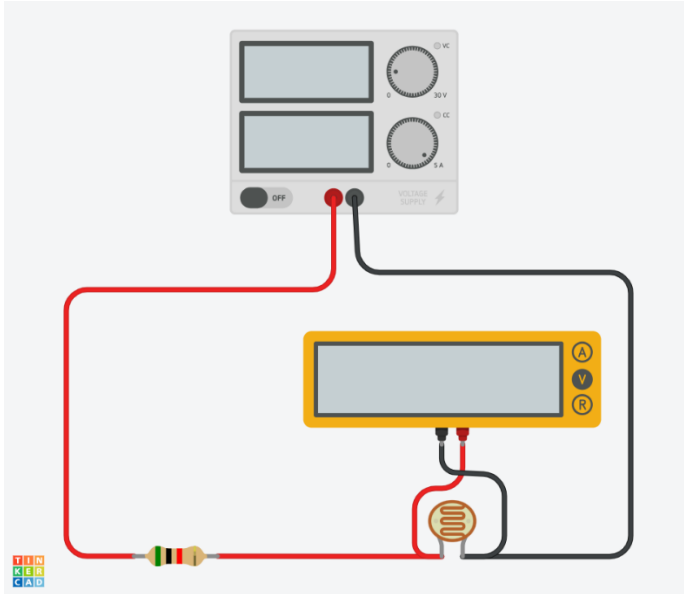


When negative voltage was connected, the diode blocked the current. When positive voltage was connected, the diode slowly started to let some current starting from the neighborhood of $V_D = 0.40\text{ V}$ to $V_D = 0.45\text{ V}$, then grows rapidly and reaches maximum at $V_D = 0.65\text{ V}$.

Part 2 Light Dependent Resistor (LDR)

A light dependent resistor (LDR) changes its resistance based on the amount of light that falls on it. A resistor ($R \approx 5\text{ k}\Omega$) and an LDR were connected in series to a power source. The light source was adjusted during the experiment to get different amounts of lighting and the resulting voltage across the LDR was measured.

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Light Dependent Resistor (LDR)	1	Semiconductors whose resistance varies on ambient light intensity
Resistor	1	$\approx 5 \text{ k}\Omega$
Connection Wires	~	Copper wires
Breadboard	1	A platform for assembling the circuit
Phone Flashlight	1	With adjustable luminosity

Variables

Name	Type	Resolution
V_{all}	Control	0.01 V
V_{LDR}	Measure	0.01 V
V_R	Calculate	0.01 V
I	Calculate	1 mA
R_{LDR}	Calculate	0.01 Ω

Errors

Name	Type	Impact
Illuminance Variance	Random	It is practically unachievable to recreate exactly same luminosity in the system without any special equipment, also the shades, angles, everything matters in this experiment.
Power Supply Fluctuations	Random	The output voltage from the DC power supply may not be perfectly stable and fluctuate slightly.
Contact and Wire Resistance	Systematic	The connecting wires and the contact points on the breadboard have a small but non-zero resistance. This can cause the total resistance to be slightly higher than the sum of the component resistors, affecting the I-V readings.

Formulas Used

$$I = \frac{V_{LDR}}{R_{LDR}} = \frac{V_R}{R}$$

$$V_R = V_{all} - V_{LDR}$$

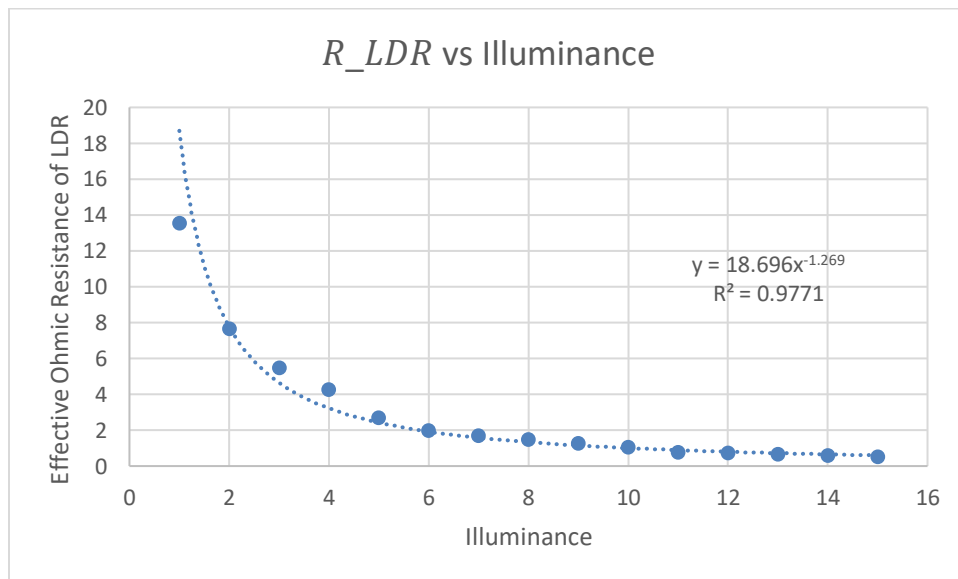
$$R_{LDR} = R \times \frac{V_{LDR}}{V_{all} - V_{LDR}}$$

Data

Illuminance	V_{LDR}	V_{all}	V_R	R_{LDR}
...	5.48	7.5	2.02	13.56436
...	4.54	7.5	2.96	7.668919
...	3.92	7.5	3.58	5.47486
...	3.46	7.5	4.04	4.282178
...	2.62	7.5	4.88	2.684426

...	2.13	7.5	5.37	1.98324
...	1.9	7.5	5.6	1.696429
...	1.72	7.5	5.78	1.487889
...	1.51	7.5	5.99	1.260434
...	1.31	7.5	6.19	1.058158
...	1.01	7.5	6.49	0.77812
...	0.945	7.5	6.555	0.720824
...	0.868	7.5	6.632	0.654403
...	0.818	7.5	6.682	0.612092
Maximal	0.69	7.5	6.81	0.506608

Plot

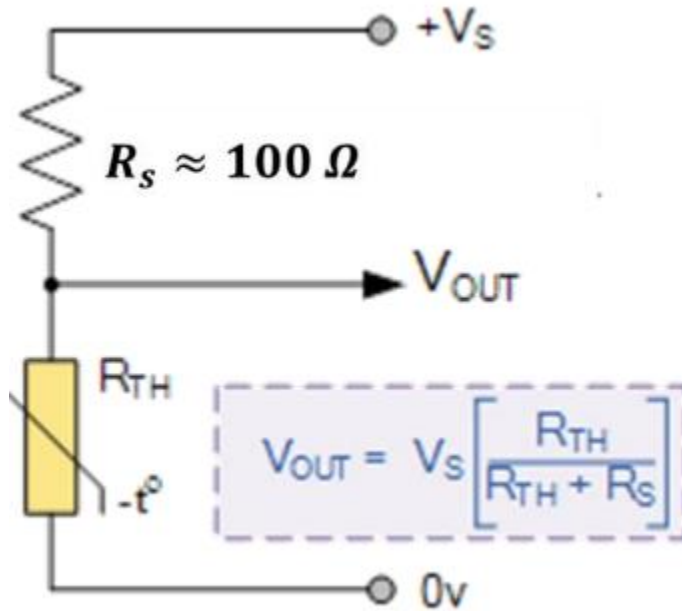


From the graph it is apparent that with increasing luminosity the resistance of the LDR drops non-linearly.

Part 3 Thermistor I-V Characteristics

The goal of this part is to measure the current-voltage (I-V) characteristics of a thermistor and analyze the effect of self-heating on its resistance. The circuit configuration utilized for this experiment is shown below:

Setup



Instruments

Name	Quantity	Properties
Power Supply	1	Generate AC/DC current through the circuit.
Multimeter	1	With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V)
Thermistor	1	Semiconductors whose resistance varies with temperature
Resistor	1	$\approx 100 \Omega$
Connection Wires	~	Copper wires
Breadboard	1	A platform for assembling the circuit

Variables

Name	Type	Units
V_{all}	Control	V
V_{TH}	Calculate	V
V_R	Measure	V
I	Calculate	mA
R_{TH}	Calculate	Ω
P_{TH}	Calculate	mW

Errors

Name	Type	Impact
Movement	Random	Any little pressure changes and wind from movement around the circuit causes the thermistor to cool down and ruin the experiment.
Power Supply Fluctuations	Random	The output voltage from the DC power supply may not be perfectly stable and fluctuate slightly.
Contact and Wire Resistance	Systematic	The connecting wires and the contact points on the breadboard have a small but non-zero resistance. This can cause the total resistance to be slightly higher than the sum of the component resistors, affecting the I-V readings.

Formulas Used

$$V_{TH} = V_{all} - V_R$$

$$I = \frac{V_R}{R}$$

$$R_{TH} = \frac{V_{TH}}{I}$$

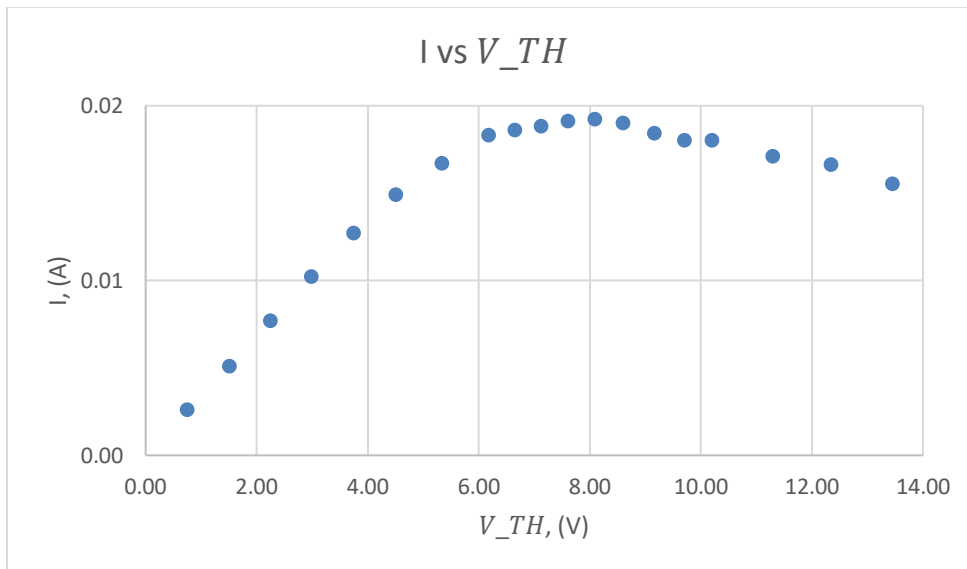
$$P_{TH} = V_{TH}I$$

Data

V_{all}	V_R	V_{TH}	I	P_{TH}	R_{TH}
1.01	0.26	0.75	0.00	0.00	288.46
2.02	0.51	1.51	0.01	0.01	296.08
3.02	0.77	2.25	0.01	0.02	292.21
4.00	1.02	2.98	0.01	0.03	292.16
5.01	1.27	3.74	0.01	0.05	294.49

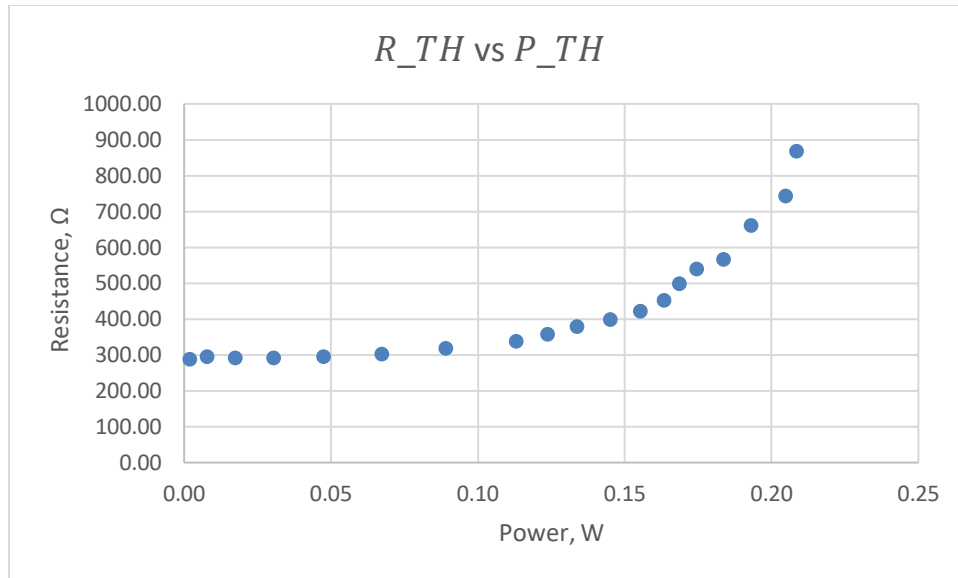
6.00	1.49	4.51	0.01	0.07	302.68
7.00	1.67	5.33	0.02	0.09	319.16
8.01	1.83	6.18	0.02	0.11	337.70
8.51	1.86	6.65	0.02	0.12	357.53
9.00	1.88	7.12	0.02	0.13	378.72
9.51	1.91	7.60	0.02	0.15	397.91
10.01	1.92	8.09	0.02	0.16	421.35
10.50	1.90	8.60	0.02	0.16	452.63
11.00	1.84	9.16	0.02	0.17	497.83
11.50	1.80	9.70	0.02	0.17	538.89
12.00	1.80	10.20	0.02	0.18	566.67
13.00	1.71	11.29	0.02	0.19	660.23
14.00	1.66	12.34	0.02	0.20	743.37
15.00	1.55	13.45	0.02	0.21	867.74

Relation between Current and Voltage of PTC Thermistor



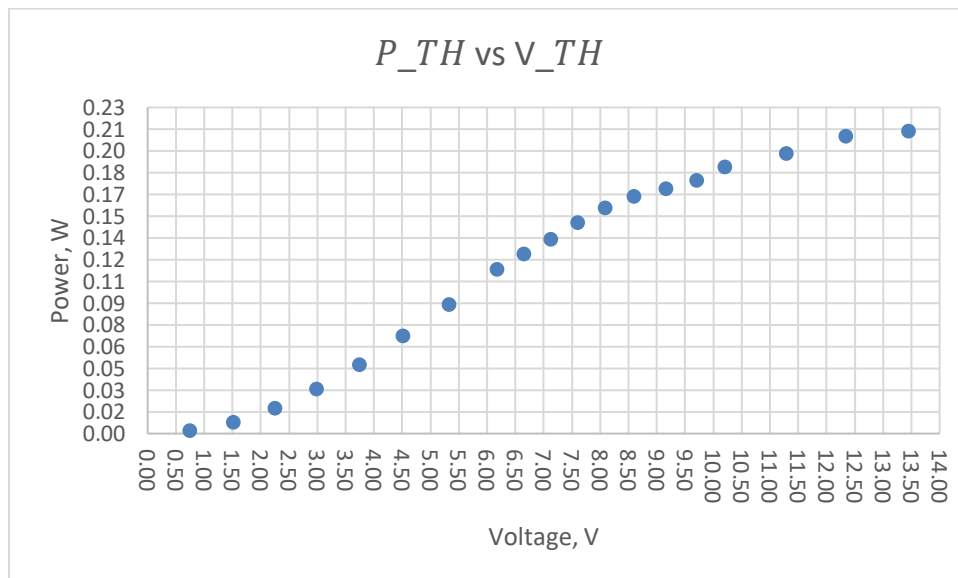
It is evident that after approximately 8 V, the current stops increasing, indicating that the peak current is reached at approximately $V_{TH} = 8$ V. It is about 0.02 A

Relation between Resistance and Power



The resistance of the thermistor remains nearly constant at its cold resistance until the power dissipated reaches approximately 0.05 W. As power increases beyond this point, the resulting self-heating raises the thermistor's temperature, causing the resistance to increase sharply.

Dependence of Power on the Voltage across the thermistor



The graph shows that the power dissipated increases non-linearly with the voltage across the thermistor. The power reaches its peak a maximum of approximately 0.21 W at a voltage of about 13.40 V.